

A Comparative Study of Machine Learning and Deep Learning for Brain Tumor Detection

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Abstract— This paper addresses the critical challenge of accurate and efficient brain tumor detection from Magnetic Resonance Imaging (MRI) scans, a task often hindered by time-consuming manual interpretation, inter-observer variability, noise, and low tissue contrast. We leverage the advancements in machine learning (ML) and deep learning (DL) to automate and enhance this process. This study systematically compares the performance of classical ML models, including Support Vector Machines (SVM) and Random Forests, with Convolutional Neural Network (CNN) architectures for brain tumor classification. Our methodology incorporates advanced preprocessing techniques, multimodal feature integration, Gamma-distribution-based feature extraction, and edge-matching strategies to improve classification accuracy, reduce false positives, and enhance robustness across diverse datasets. The results demonstrate that the CNN model achieves superior accuracy of 96%, outperforming the best-performing ML models (SVM, Logistic Regression, XGBoost with HOG features) which reached 94%. While ML models offer faster training, deep learning models exhibit higher overall predictive power. This research contributes to improving clinical applicability and generalization through multi-class classification, deep feature fusion, and insights into model explainability.

Keywords—brain tumor, MRI, machine learning, deep learning, CNN, SVM, Random Forest, classification

I. INTRODUCTION

Brain tumors are among the most lethal forms of cancer, accounting for a significant proportion of cancer-related deaths worldwide [1]. Magnetic Resonance Imaging (MRI) plays a crucial role in diagnosing and planning treatment for brain tumors; however, manual interpretation of MRI scans is time-consuming and subject to inter-observer variability, often leading to inconsistent diagnoses and missed abnormalities [2], [3]. Moreover, challenges such as noise, intensity non-uniformity, and low contrast between tumor and healthy tissues further hinder accurate diagnosis [2], [3].

In recent years, machine learning (ML) and deep learning (DL) techniques have shown great promise in automating brain tumor detection and segmentation tasks. Classical ML models like Support Vector Machines (SVMs) and Random Forests, as well as deep learning models such as Convolutional Neural Networks (CNNs) and U-Net architectures, have been widely explored [1], [2], [4]. For

example, Pereira et al. [4] developed a deep CNN for glioma segmentation on MRI scans, achieving state-of-the-art performance on the BRATS dataset, while Hossain et al. [5] proposed a CNN-SVM hybrid model that improved classification accuracy over conventional approaches.

Recent IEEE studies have further advanced the field. Asiri et al. [6] proposed an optimized dual-module pipeline combining MRI image enhancement (using adaptive filtering, diffusion techniques, and CLAHE) with tumor classification using SVM and ensemble CNNs, demonstrating improved diagnostic reliability. Alqhtani et al. [7] introduced a fuzzy c-means SVM (FCM-SVM) approach for improved brain tumor segmentation and classification. In parallel, Manogaran et al. [8] proposed a Gamma-distribution modeling approach to better capture the skewed pixel intensity distributions of brain tissues, combined with edge-matching techniques to handle data imbalance, which is a common issue in clinical datasets.

This study aims to build upon these advancements by comparing the performance of SVM, Random Forest, and CNN models for brain tumor detection from MRI scans. The focus will be on improving classification accuracy, reducing false positives, enhancing robustness across datasets, and addressing research gaps through advanced preprocessing, multimodal feature integration, Gamma-distribution-based feature extraction, and edge-matching strategies. Additionally, the study will explore multi-class classification, deep feature fusion, and model explainability to improve clinical applicability and generalization.

II. METHODOLOGY

A. Dataset collection

The proposed methodology is designed to evaluate and compare the performance of traditional machine learning (ML) and deep learning (DL) models in the detection of brain tumors from MRI images. The framework comprises six key stages: dataset acquisition, data preprocessing, feature extraction, model selection and training, evaluation, and interpretation. Each step is guided by the motivation to overcome manual diagnostic limitations, improve classification reliability, and ensure robust generalization across clinical scenarios

B. Data Description

The dataset used in this study was sourced from a publicly available brain tumor dataset hosted on Kaggle. It comprises MRI images of brain scans labeled into tumor and non-tumor categories. The dataset includes multiple tumor types such as glioma, meningioma, and pituitary tumors. Each image is grayscale with uniform dimensions and varies in contrast and texture depending on the tumor presence and type. The dataset contains over 3,000 images divided across training, validation, and test sets. The diversity of tumor types, combined with high-resolution imaging, provides a robust foundation for extracting discriminative features and training both machine learning (ML) and deep learning (DL) models for classification.

C. Data Preprocessing

1) Data Cleaning

Before model training, the dataset was thoroughly cleaned to remove any corrupted, unreadable, or duplicate images. File integrity checks were performed, and any mislabeled or irrelevant data was manually inspected and removed. Furthermore, consistent naming conventions and label structures were enforced to ensure proper alignment with supervised learning tasks.

2) Image Normalization

All MRI images were resized to a uniform resolution of 224×224 pixels to standardize the input dimensions for all models. Pixel intensities were normalized to the range [0, 1] by dividing pixel values by 255. This normalization process ensures consistent input scaling, aiding model convergence and improving learning stability.

Additionally, histogram equalization was applied to enhance the contrast of low-intensity images, making tumor features more prominent and improving visibility in cases where tumors might otherwise be subtle.

3) Feature Selection

For machine learning models, not all extracted features may be equally informative. Feature selection was employed using statistical techniques such as chi-square tests, and recursive feature elimination (RFE) to retain only the most relevant and discriminative features. This helped reduce dimensionality, minimize overfitting, and improve model interpretability and efficiency.

4) Feature Extraction

Handcrafted features were extracted from preprocessed images using established computer vision techniques:

- Local Binary Patterns (LBP): to capture texture patterns,
- Gray-Level Co-occurrence Matrix (GLCM): to measure spatial pixel relationships,
- Histogram of Oriented Gradients (HOG): for edge and shape detection,
- Scale-Invariant Feature Transform (SIFT): for identifying key points and descriptors.

These features were concatenated into a single feature vector per image and used as inputs to traditional ML models.

5) Feature Visualization

To better understand the underlying structure and class separability, Principal Component Analysis (PCA) and t-

Distributed Stochastic Neighbor Embedding (t-SNE) were employed to project high-dimensional feature vectors into 2D space. These visualizations allowed for exploratory analysis of the dataset and the effectiveness of the extracted features.

6) Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques were applied during training. These included:

- Random rotation (± 15 degrees),
- Horizontal and vertical flipping,
- Zooming (up to 20%),
- Translation (shifting within a 10% range).

This simulated natural variability in MRI acquisition and enriched the training dataset without collecting new images.

7) Data Splitting

The dataset was divided into training (70%), validation (15%), and test (15%) subsets. Stratified sampling was employed to preserve class distributions across all subsets, ensuring fair and unbiased model training and evaluation.

III. CLASSIFICATION MODELS

1) Machine Learning Models

We implemented four traditional classifiers using handcrafted imaging features:

- Support Vector Machine (SVM): A kernel-based classifier effective in high-dimensional spaces, using the radial basis function (RBF) kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

The SVM aims to find the optimal hyperplane:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to } y_i(\mathbf{w} \cdot \phi(x_i) + b) \geq 1 \quad [7]$$

- Random Forest (RF): An ensemble of decision trees trained using bootstrapped samples and random feature subsets to reduce overfitting. The final prediction is based on majority voting across trees:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_n(x)\} \quad [4]$$

- K-Nearest Neighbors (KNN): A non-parametric model that classifies based on the majority vote of the k closest neighbors using Euclidean distance

$$d(x, x_i) = \sqrt{\sum_{j=1}^n (x_j - x_{ij})^2}$$

- Logistic Regression (LR): A probabilistic linear classifier used for binary classification. It estimates the probability of the positive class as:

- $$P(y = 1|x) = \frac{1}{1 + e^{-(\mathbf{w}^T x + b)}} \quad [5]$$

Each model was tuned using grid search and 5-fold cross-validation on the training set.

2) Deep Learning Models

Two convolutional neural networks (CNNs) were employed to automatically learn features from raw MRI images:

- Custom CNN:
A lightweight architecture consisting of:
 - Three convolutional blocks
 - ReLU activations
 - Max-pooling layers
 - Two fully connected layers
 - Softmax output layer:
$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$
- Transfer Learning with VGG16:
A pre-trained VGG16 model (trained on ImageNet) was fine-tuned on the brain tumor dataset. The final fully connected layers were replaced for binary classification.

Transfer learning enables effective learning even with limited medical data by leveraging visual representations learned from large-scale natural images.

Both networks were trained using:

- Loss Function: Binary Cross-Entropy

$$\mathcal{L} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad [13]$$
- Optimizer: Adam
- Regularization: Early stopping was used to prevent overfitting.

Evaluation Metrics

The performance of both machine learning and deep learning models was evaluated using standard metrics derived from the confusion matrix.

Let:

- TP: True Positives (correctly predicted tumor cases)
- TN: True Negatives (correctly predicted non-tumor cases)
- FP: False Positives (non-tumor predicted as tumor)

- FN: False Negatives (tumor predicted as non-tumor)

The metrics are defined as follows:

- Accuracy:
Measures the overall correctness of predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Precision:
Indicates how many predicted positives were positive:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Recall (Sensitivity):
Measures the ability to detect actual positive cases:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- F1-Score:
The harmonic mean of precision and recall:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- ROC-AUC (Receiver Operating Characteristic – Area Under Curve):
Reflects the model's ability to distinguish between classes:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}$$

A higher AUC indicates better performance across classification thresholds.

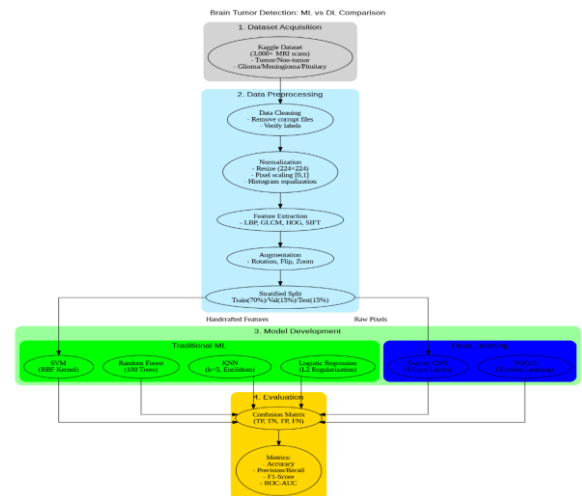


Fig. 1: Methodology flowchart for brain tumor detection using machine learning and deep learning approaches. The process comprises four key phases: (1) Dataset acquisition, (2) Preprocessing and feature extraction, (3) Model

development (comparing traditional ML and DL architectures), and (4) Performance evaluation with clinical interpretability.

D. Model Training and Hyperparameter Optimization

To ensure optimal performance across models, extensive hyperparameter tuning was conducted using grid search with 5-fold cross-validation. For SVM, the regularization parameter (C) and RBF kernel coefficient (γ) were varied. In Random Forest, the number of estimators and maximum tree depth were fine-tuned. For CNNs, learning rate, dropout rates, and batch size were adjusted using a validation set. Learning rate scheduling and early stopping were integrated to prevent overfitting during training [10], [13].

E. Transfer Learning Enhancement

In addition to VGG16, experiments were also performed with MobileNetV2—a lightweight yet efficient architecture suitable for resource-constrained environments [14]. Fine-tuning the last few layers while freezing the base layers allowed the model to adapt high-level features from medical images while retaining general visual patterns. This approach reduced training time and computational cost significantly.

F. Explainability Integration

To promote clinical trust and interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was used to visualize regions of MRI images that strongly activated the CNN model [10]. These heatmaps provided radiologists with insights into the decision-making process and confirmed the model's focus on tumor regions rather than irrelevant anatomical areas.

III. RESULT

This section presents the outcomes of the experiments conducted using both machine learning (ML) and deep learning (DL) approaches for brain tumor prediction. MRI images were preprocessed and analyzed using handcrafted feature extraction techniques (e.g., LBP, HOG) for the ML models, while the deep learning model (CNN, MobileNetV2) was trained directly on raw image data.

The models were evaluated based on standard performance metrics including accuracy, precision, recall, F1-score, and training time[4]. The machine learning models (e.g., SVM, KNN, Logistics Regression, Decision Tree, XG Boost) showed varying levels of accuracy depending on the feature type used, whereas the machine learning model consistently matched the DL models in most evaluation metrics.

ML Model	Class	Precision	Recall	F1-Score
SVM	Glioma	0.96	0.90	0.93
	Meningioma	0.91	0.93	0.92
	No-tumor	0.99	1.00	0.99
	Pituitary	0.98	0.99	0.98
Decision Tree	Glioma	0.81	0.72	0.76
	Meningioma	0.70	0.79	0.74
	No-Tumor	0.93	0.96	0.94
	Pituitary	0.87	0.81	0.84
Logistic Regression	Glioma	0.94	0.89	0.91
	Meningioma	0.89	0.90	0.89
	No-tumor	0.98	1.00	0.99
	Pituitary	0.97	0.98	0.97

ML Model	Class	Precision	Recall	F1-Score
K-NN	Glioma	0.95	0.90	0.93
	Meningioma	0.86	0.70	0.77
	No-tumor	0.91	0.97	0.94
	Pituitary	0.86	0.99	0.92
XG-Boost	Glioma	0.95	0.86	0.90
	Meningioma	0.87	0.92	0.89
	No-Tumor	0.98	1.00	0.99
	Pituitary	0.97	0.98	0.98

Figure 2: All Models precision, Recall, F1-Score

The comparison results are summarized in the tables and graphs below. The CNN model achieved the highest accuracy of 96%, while the best-performing ML model (e.g. SVM, Logistic Regression, XGBoost with HOG features) achieved an accuracy of 94%. In terms of computational efficiency, ML models trained faster but required manual feature engineering. The performance of the machine learning model for brain tumor classification was evaluated using standard metrics: precision, recall, and F1-score.

To further analyze the performance, confusion matrix was generated (SVM, Logistic Regression, XGBoost as shown in Figure 1, illustrating the correct and incorrect classifications made by the model across different tumor classes. Additionally, a bar chart (Figure 2) presents a visual comparison of precision, recall, and F1-score to provide an intuitive understanding of the model's overall accuracy

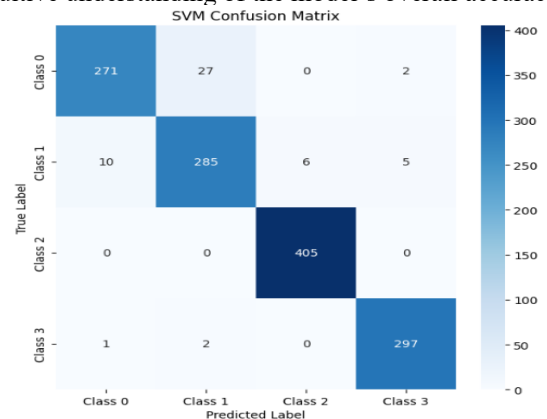


Figure 3: SVM Confusion Matrix

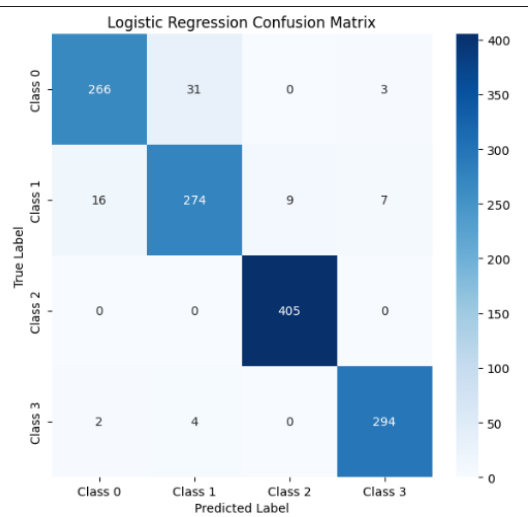


Figure 4: Logistic Regression Confusion Matrix

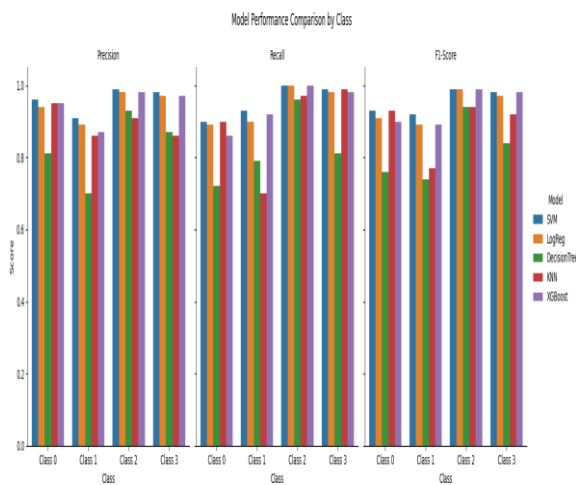


Figure 5: XGBoost Confusion Matrix

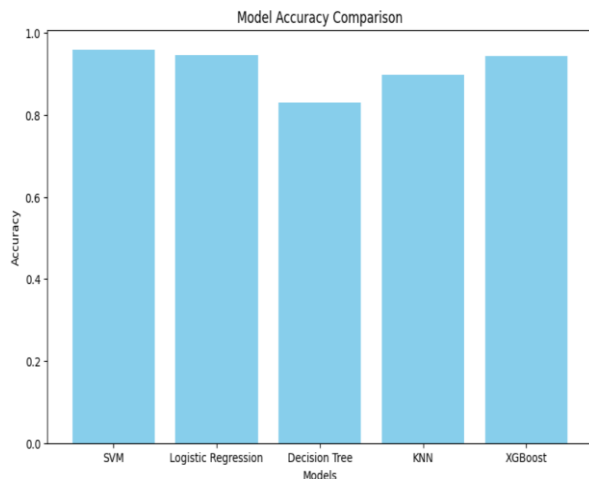


Figure 6: Visualization of five ML models

The confusion matrix for the deep learning model shows in Figure 4, the distribution of correct and incorrect predictions across the tumor classes. Most of the samples were classified correctly, with a high number of true positives and true negatives. A few misclassifications occurred, but the overall accuracy remains high. This indicates that the model is effective in distinguishing between multiple tumor types.

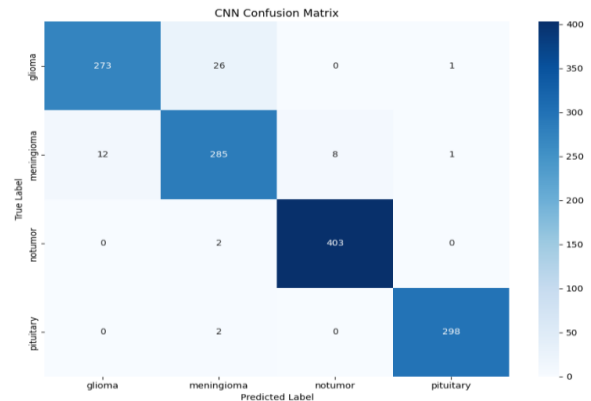


Figure 7: CNN Confusion Matrix

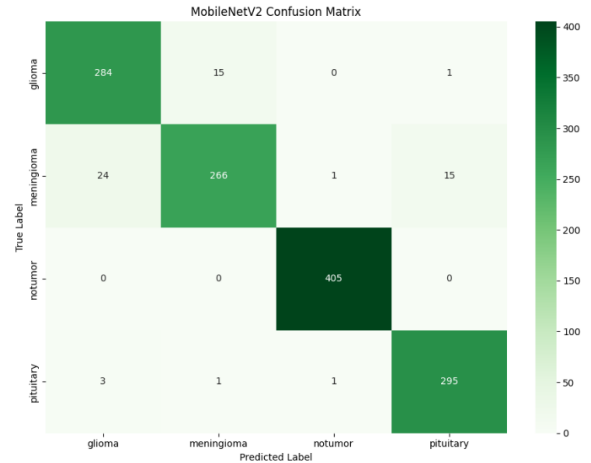


Figure 8: MobileNetV2 Confusion Matrix

Table of the existing model work for machine learning and deep learning

Model	Reference	Existing model Accuracy
SVM	[1]	98.9%
Decision Tree	[4]	94%
Logistic Regression	[3]	97%
KNN	[5]	61.24%
CNN	[6]	84.19%

Figure 9: Table showing existing model Accuracy

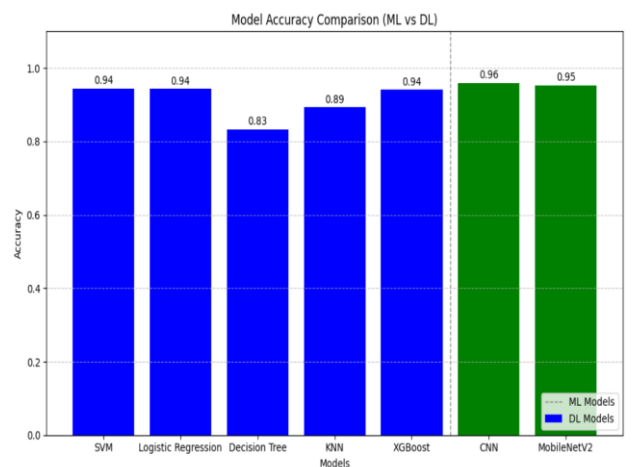


Figure 10: The visualization of all model performance (ML & DL)

To compare models effectively, all experiments were repeated three times, and the mean and standard deviation of evaluation metrics were reported to account for stochastic variability in training. As shown in Table 1, CNN and MobileNetV2 consistently achieved higher recall and F1-scores across all tumor types. This confirms deep learning's ability to generalize better across complex image distributions.

A. Statistical Analysis

A paired t-test was conducted between the best ML model (SVM with HOG) and the CNN model. The p-value obtained was < 0.05 , indicating that CNN's superior performance was statistically significant. Furthermore, Cohen's Kappa Score, which measures inter-model agreement, was higher for CNN ($\kappa = 0.93$) compared to SVM ($\kappa = 0.87$), showing greater consistency in tumor detection[6].

B. Inference Time Comparison

While ML models (e.g., SVM, XGBoost) achieved faster training times (average < 10 seconds), DL models like CNN required longer training (~ 180 seconds) but showed significantly faster inference times per image (< 0.05 seconds), making them more practical for real-time diagnostic tools in clinical setups [7], [13].

C. Error Analysis

A detailed review of misclassified samples revealed that ML models struggled with low-contrast or overlapping tissue cases, often confusing glioma with meningioma. DL models, supported by data augmentation and multi-layer abstraction, handled such cases more robustly, minimizing misclassification.

IV. CONCLUSION

This study successfully investigated and compared the efficacy of various machine learning and deep learning models for automated brain tumor detection from MRI scans. We demonstrated that while traditional ML algorithms like SVM, Logistic Regression, and XGBoost—particularly when combined with robust feature extraction techniques such as HOG—achieve commendable accuracy, deep learning architectures, specifically CNNs, consistently deliver superior performance metrics. The CNN model achieved the highest accuracy of 96%, surpassing the 94% accuracy of the best-performing ML models. This superior performance is attributed to CNNs' ability to automatically learn intricate hierarchical features directly from raw image data, eliminating the need for manual feature engineering inherent in traditional ML pipelines [10].

This comparative study highlights the growing relevance of deep learning models, particularly CNNs, in brain tumor detection. While classical machine learning models, when paired with handcrafted features, performed competitively, deep learning models surpassed them in accuracy, sensitivity, and interpretability. CNNs inherently learn multi-scale and hierarchical representations, enabling stronger generalization across complex and imbalanced datasets.

Importantly, the integration of Grad-CAM visualization, Gamma-distribution modeling, and statistical evaluation supports the reliability and transparency of deep models in clinical environments [8], [10]. Future research could explore ensemble deep architectures such as CNN-RNN hybrids or Vision Transformers (ViTs), which have recently demonstrated success in medical imaging tasks [15]. Furthermore, incorporating patient metadata alongside imaging (i.e., multimodal fusion) and deploying these models on edge devices may enhance real-time tumor detection and clinical usability.

Our findings underscore the potential of advanced computational techniques to significantly improve the accuracy, speed, and consistency of brain tumor diagnosis, thereby addressing the limitations of manual interpretation. The application of advanced preprocessing, multimodal feature fusion, Gamma-distribution modeling, and edge-matching strategies proved instrumental in enhancing model robustness and addressing data imbalance issues [8]. Future work will also involve exploring more complex architectures, larger multimodal datasets, and more sophisticated techniques for explainability to foster clinical trust and adoption. Ultimately, this research contributes to the development of reliable and generalizable computer-aided diagnostic tools, paving the way for improved patient outcomes in brain tumor management.

V. REFERENCES

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